The empirical recurrence for OEIS A237941: recovering a conjecture that is not written down

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Abstract

OEIS A237941 records "Empirical recurrence of order 54", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 345$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 54, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 54 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A237941 is "Number of (n+1)X(4+1) 0..2 arrays with the maximum minus the minimum of every 2X2 subblock equal.". It has offset 1 and begins

$$13893, 946063, 61286941, 4202495253, 287601819479, 19789034062007, 1361783640641509, 93760075445980377, \ldots$$

The entry states:

> Empirical recurrence of order 54 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 10:37:07, revision 7) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 345$ vertices, $a(n) = \iota^{\top} M^{j(n)} \tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' =$ 345.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 690$ exact terms of the model returns order 54 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{54} c_i\, a(n-i),$$

c_1	114	c_{15}	110002571126724549	c_{29}	938212562586266131806940	c_{43}	2016336
c_2	−2867	c_{16}	−829556868674006798	c_{30}	−959939086000441677257788	c_{44}	−855816
c_3	−52231	c_{17}	−1130920306950853929	c_{31}	−2038816696572809784140896	c_{45}	−303435
c_4	2711827	c_{18}	18146530385042371408	c_{32}	3609791640488783037461504	c_{46}	257929
c_5	−8661804	c_{19}	−6950167137661256282	c_{33}	2220416381017563009980720	c_{47}	−84048
c_6	−757325894	c_{20}	−250068856733869338602	c_{34}	−7837086690644302787241200	c_{48}	−30686
c_7	7527306920	c_{21}	386266227835423261411	c_{35}	610276958764461994800416	c_{49}	59675
c_8	79807056462	c_{22}	2190664090388691523297	c_{36}	10303905957021762786240576	c_{50}	13229
c_9	−1367998363447	c_{23}	−5761584646668520683269	c_{37}	−5548883776251089714762048	c_{51}	−4774
c_{10}	−1687663009317	c_{24}	−11442161936878462930223	c_{38}	−7713746166965585532004736	c_{52}	3181
c_{11}	110917108096881	c_{25}	49132704115136727028958	c_{39}	7880574088008778526416384	c_{53}	1138
c_{12}	−258739310124081	c_{26}	24394672643387282353612	c_{40}	2354164560770143102939648	c_{54}	−994
c_{13}	−4739655400616131	c_{27}	−266910375387161278937832	c_{41}	−5548434853033827411120128		
c_{14}	22152458780958445	c_{28}	97283434548590124500248	c_{42}	757783782272487607109632		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 54, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $54+4$ terms removed returns order 54 as well, so $\deg q_{\min} = 54 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 12 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $54+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 54 that this sequence satisfies — and that family turns out to have exactly one member.

References

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